



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION 2

CANDIDATE
NAME

CENTRE
NUMBER

INDEX
NUMBER

CLASS

H2 BIOLOGY

9648/02

Paper 2 Core Paper

16 September 2015

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Answer any **one** question on the answer paper provided.

Circle the question attempted on the cover page.

The use of an approved scientific calculator is expected, where appropriate.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 13
2	/ 9
3	/ 11
4	/ 8
5	/ 9
6	/ 10
7	/ 10
8	/ 10
Section B	
9 or 10*	/ 20
Total	

This Question Paper consists of **20** printed pages.

Section A (80 marks)

Answer **all** the questions in this section.

1. Fig 1.1 is an electron microgram showing part of the cell body of a neurone.

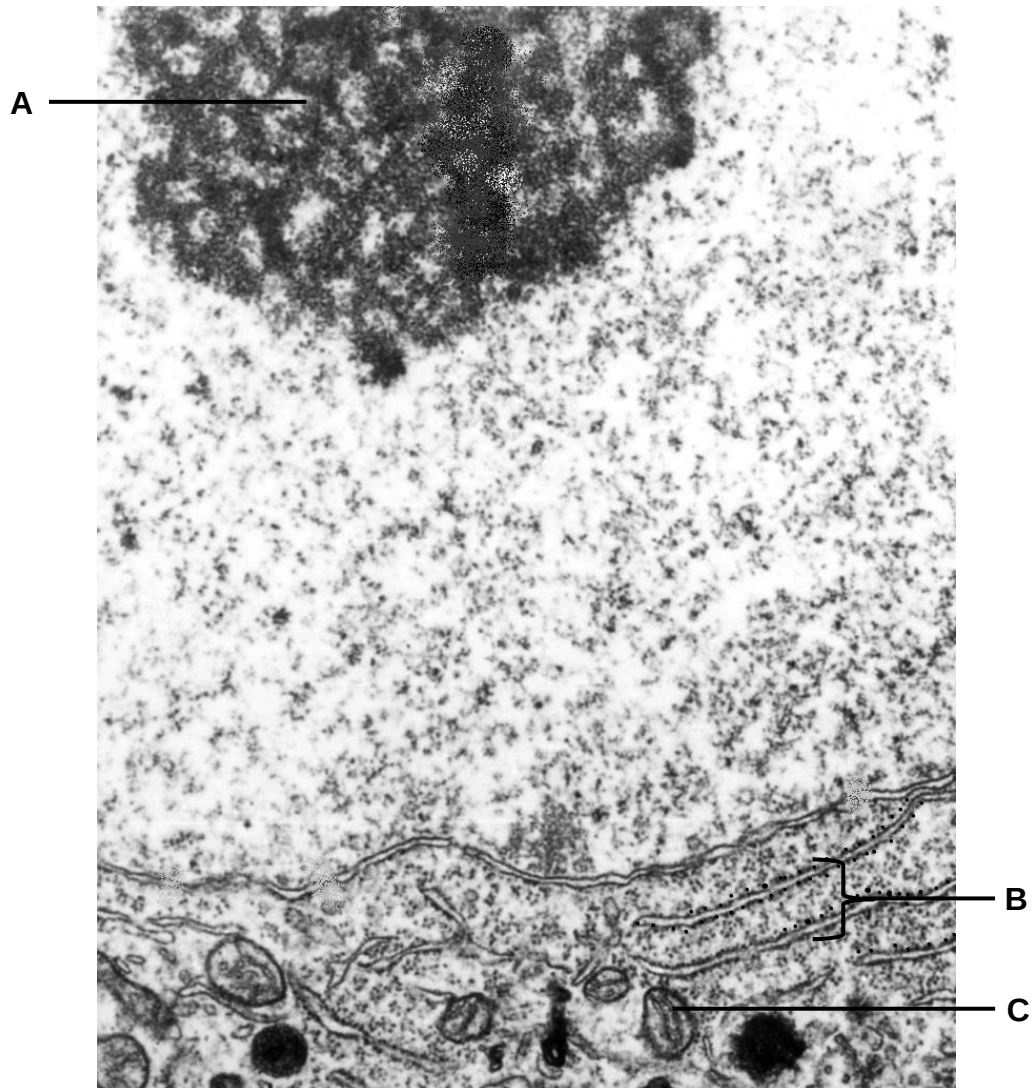


Fig 1.1

(adapted from http://www.columbia.edu/itc/hs/medical/sbpm_histology_old/lab/micro_popup15.html)

- (a) Name the structures labelled **A** and **B**. [3]

A Nucleolus;;

B Rough Endoplasmic Reticulum;;

C Mitochondrion

Dopamine is a neurotransmitter associated with the 'pleasure system' of the brain, providing feelings of enjoyment and reinforcement to motivate us to do, or continue doing certain activities.

Dopamine is synthesised in dopaminergic neurones via a multi-step process involving enzymes tyrosine hydroxylase and aromatic amino acid decarboxylase (AAAD).

An outline of the biochemical pathway for the synthesis of dopamine is shown in Fig 1.2.



Fig 1.2

(b) With reference to Fig 1.1 and 1.2, suggest the role(s) of structures **A**, **B** and **C** in the synthesis of dopamine. [4]

1. **A** is the site of transcription of rRNA genes;
2. site of assembly of ribosomal subunits;
3. for ribosomes required for the translation of mRNA of tyrosine hydroxylase and AAAD;
4. **B** is the site of protein synthesis;
5. where mRNA are translated into polypeptide of enzymes;
6. and folded to specific three dimensional conformation;
7. **C** is the site of ATP production during aerobic respiration;
8. to provide energy for peptide bond formation during translation;

Fig 1.3 shows a drawing of a synapse between two dopaminergic neurons.

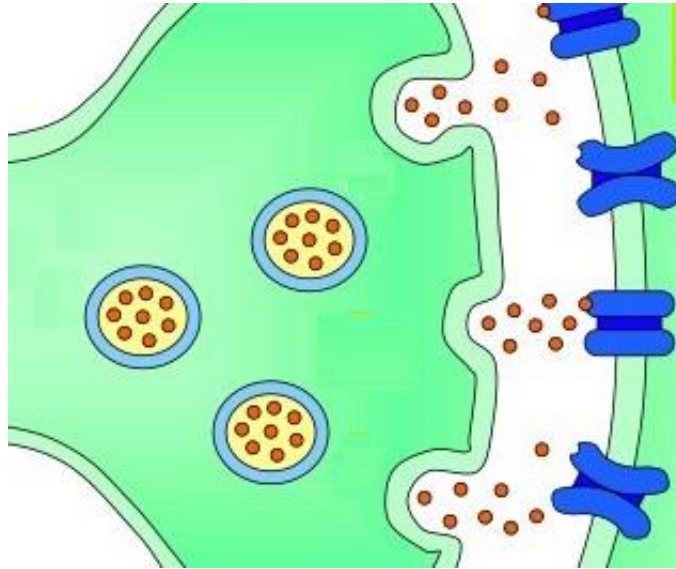


Fig 1.3

(c) Describe the processes shown in Fig 1.3, which leads to the depolarisation of the post-synaptic membrane. [4]

1. Membrane of secretory vesicles containing dopamine fuse with the pre-synaptic membrane;
2. releasing dopamine into the synaptic cleft;
3. via exocytosis;
4. Dopamine diffuse across synaptic cleft;
5. binds to dopamine receptors on chemically-gated sodium channels;
6. upon binding of dopamine, dopamine receptor undergoes conformational change/opening of sodium channels;
7. allow passage of positively charged sodium ions/influx of sodium ions;
8. down it's concentration gradient;

Excessive neurotransmission of dopamine is associated with schizophrenia, a clinical condition marked by disordered thoughts.

Neuroleptics are a class of compounds with a high affinity for dopamine receptors and are

used to alleviate symptoms of the condition.

(d) Suggest how neuroleptics work to alleviate symptoms of schizophrenia. [2]

1. Neuroleptics bind to dopamine receptors on post synaptic membrane;
2. but does not trigger opening of chemically gated sodium channels;
3. Dopamine unable to bind to dopamine binding site of specific receptor;
4. No action potential initiated at post synaptic membrane;

[Total: 13]

2. Fig. 2.1 is a photomicrograph of a root tip of *Allium cepa*.

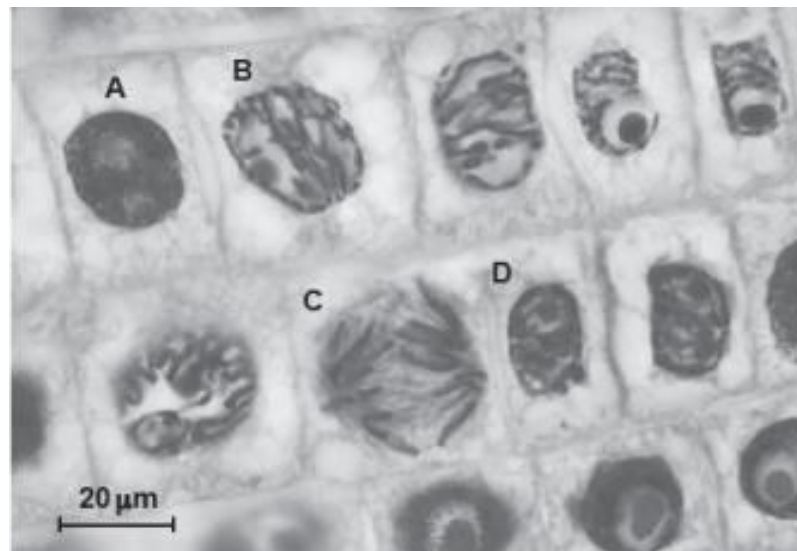


Fig. 2.1

(a) Describe the events that occur during stage **B**. [2]

1. Prophase;
2. chromatin condense / coil up / shorten;
3. nucleolus disappears;
4. nuclear envelope, breaks down / disintegrates;
5. spindle fibres formation / assembly of microtubules;

(b) Explain the significance of stage **C** in a growing plant root tip. [3]

1. Centromere separates;
2. genetically identical sister chromatids separate;
3. are pulled to opposite poles of cell;
4. producing genetically identical / same, number / type, of chromosomes; Accept: remain diploid;;
5. perform same function as part of the same tissue;;
6. repair / replacement (of root tip / tissue) or producing more cells
Reject: repair of cells;;

(c) Distinguish between the terms *sister chromatids* and *homologous chromosomes*. [2]

1. Sister chromatids have the same gene loci and same alleles along their length; but homologous chromosomes have the same gene loci but can be occupied by identical or different alleles;
2. Sister chromatids are joined together by the centromere; but the members of each pair are not attached at the centromere;

In normal cells, the cell cycle is controlled at checkpoints. Fig. 2.2 shows the main stages in the cell cycle and the three checkpoints

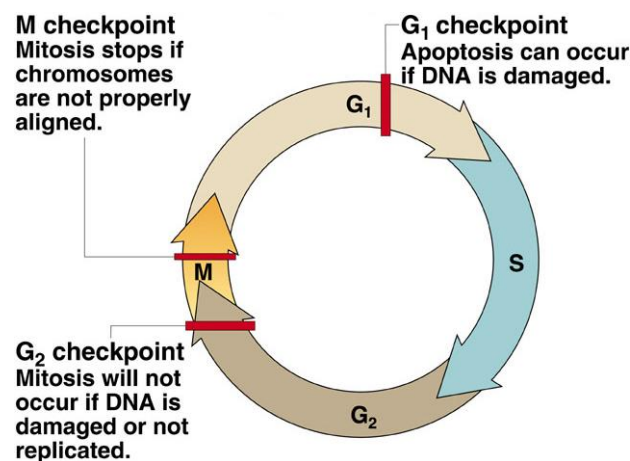


Fig. 2.2

(d) With reference to Fig. 2.2, suggest the consequence of dysregulation of **M** checkpoint. [2]

1. Proceed to anaphase, even when chromosomes are not aligned at the metaphase plate;;

2. Chromosomes being separated unequally into 2 daughter cells / non-disjunction occurs; during anaphase;
3. Results in polyploidy/aneuploidy;

[Total: 9]

3. Fig. 3.1 is an electron micrograph showing the process of protein synthesis in a prokaryote.

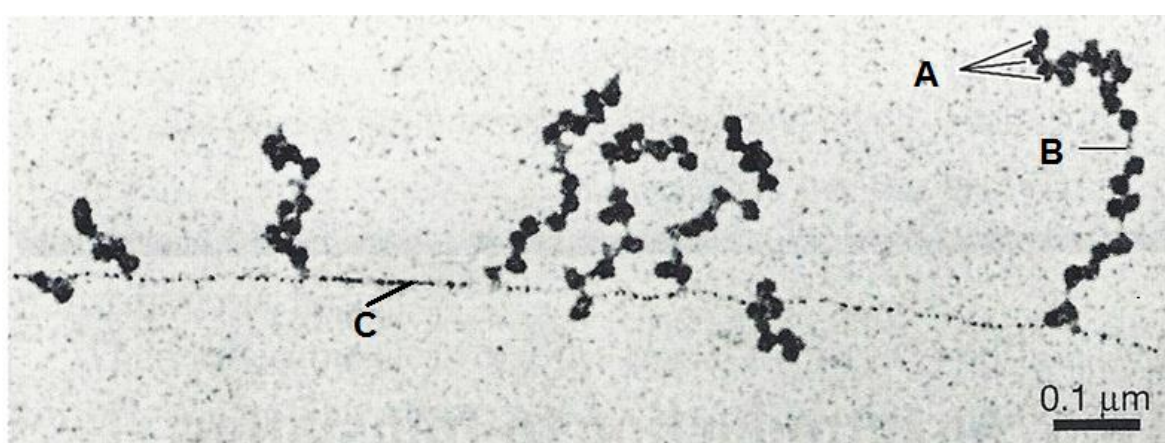


Fig. 3.1

- (a) Identify structures A and C. [2]

A Ribosomes;;

C Deoxyribonucleic acid;;

- (b) Describe how structure B is synthesised in prokaryotes. [3]

1. RNA polymerase recognises and binds to the promoter of the gene;
2. causing the DNA double helix to unwind and separate;
3. 1 of the two DNA strands serve as template strand;
4. Free ribonucleoside triphosphates formed complementary base pairing with ribonucleotides on the template DNA strand;
5. RNA polymerase catalyses formation of phosphoester bonds between ribonucleotides through condensation reaction;

6. Transcription proceeds until after the RNA polymerase transcribes a termination sequence;

(c) Describe how structure **A** is adapted to its function in protein synthesis. [2]

1. Comprises of one binding site for mRNA;
2. three binding sites for the tRNAs / A-site, the P-site and E-site;
3. binding sites are positioned close together;
4. so that mRNA and tRNA can be brought together in the process of translation;
5. contains peptidyl transferase;
6. catalyses formation of peptide bond between amino acids;

(d) With reference to Fig 3.1, describe and explain how the pattern of protein synthesis is different from that in eukaryotes. [2]

1. Transcription and translation occur simultaneously in prokaryotes but transcription occurs in nucleus and translation in cytosol/RER in eukaryotes;;
2. Due to the lack of nuclear envelope;;

(e) Comparing to prokaryotic cells, suggest why only about 1% of genetic information is transcribed into functional RNA sequences in most eukaryotic cells. [2]

1. due to differential gene expression/cell specialization/differentiation; only some genes are transcribed and translated while other genes that encode proteins not required by the cell are turned off;
2. presence of non-coding sequences such as centromeres/ telomeres/ regulatory sequences, which are not transcribed into functional RNA sequences;;

[Total: 11]

4. The HIV virus was first discovered in 1984. It was found to be able to infect human cells involved in the immune system.

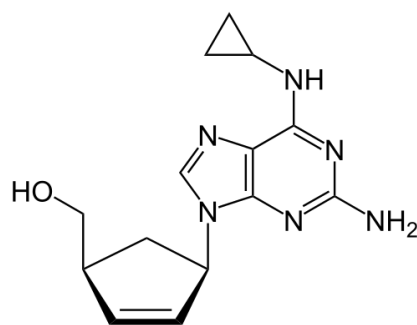
(a) Describe how HIV enters a host cell.

[3]

1. Gp120 binds to CD4 receptor on;
2. host cell surface membrane;
3. gp120 undergoes conformational change;
4. enabling gp120 to bind to co-receptors on host cell membrane (CXCR4/CCR5);
5. Interaction between gp120 and co-receptor brings about conformational change to gp41 on viral envelope;
6. leads to fusion of HIV envelope to the host cell surface membrane, gaining entry;

With modern medicine, HIV-positive patients can now live longer life. Highly Active Antiretroviral Therapy (HAART) whereby patients take two or more anti-retrovirus drug concurrently has been proven to be an effective therapy for HIV patients. One of the drugs known to be used in HAART is Abacavir.

Fig 4.1 shows the structure of Abacavir, which is a nucleoside reverse transcriptase inhibitor.



Abacavir

(b) Suggest how Abacavir can prevent HIV. [3]

1. It has shape similar to (purine) nucleoside triphosphate;
2. can fit into active site of reverse transcriptase;
3. and incorporated into viral DNA;
4. resulting in chain termination;
5. shorter DNA produced/DNA produced may lack complete gene sequences;
6. leads to no or non-functional viral proteins
7. unable to infect other host cell;

Or

3. Prevents deoxyribonucleoside triphosphate / viral ssRNA from binding to active site
4. no cDNA produced
5. cDNA of HIV not integrated into host genome/no provirus formation;
6. leads to no viral proteins produced

(c) State two ways in which the life cycle of lambda bacteriophage is different from HIV? [2]

lamda bacteriophage	HIV
Capsid does not enter bacteria host cell;	Nucleocapsid/capsid enters host cell;

Viral genome does not need to undergo reverse transcription;	ssRNA HIV genome reverse transcribed before integration into host genome;
No polyproteins are produced;	Viral gene transcribed and translated into polyproteins;
Maturation before release;	Maturation after release;
AVP	AVP

1 mark per comparison, Any two

[Total: 8]

5. Mutations in the *BRCA1*, which is a tumour suppressor gene, are associated with an increased risk of breast cancer. Researchers have identified hundreds of mutations in the gene. Women with *BRCA1* gene mutations are associated to an 80% risk of developing breast cancer by age 90 and increased risk of developing ovarian cancer.

BRCA1 gene encode a nuclear protein in human. BRCA1 protein contains two functional domains: an N-terminal Ring domain and a C-terminal BRCT domain. The C-terminal BRCT domain of BRCA1 is a DNA binding domain, which is also required for BRCA1's translocation and accumulation at DNA damage sites for DNA repair.

The three alleles of *BRCA1* gene have the following mutations in C-terminal BRCT domain:

- single-base substitution at nucleotide 1047
- single-base insertion at nucleotide 5382
- single-base deletion at nucleotide 6174

- (a) (i) Identify and explain which mutation(s) has the most serious consequences for the structure of the BRCA1 protein. [3]

1. single-base insertion; and

2. single-base deletion;
3. frame shift mutation/ alters reading frame (after mutation);
4. (so) all amino acids different after mutation; A: stop codon
5. 3-D conformation/tertiary structure, of protein changed;
A: truncated polypeptide chain
6. (whereas) substitution, may change only one amino acid/may be silent;

(ii) Explain how the change in (a)(i) could result in cancer. [2]

1. Change in shape of DNA binding domain;
2. no longer complementary to shape of DNA;
3. BRCA1 protein cannot translocate and accumulate at DNA damage sites (for repair);
4. accumulation of mutations leading to uncontrolled cell division;

The *HER2* gene is another gene that link to breast cancer. HER2 protein is a growth factor receptor on the surface of normal breast cells. When the growth factor binds to HER2 protein, the cell is stimulated to grow and divide. Gene amplification of *HER2* gene is found in cancer cells.

Breast cancer targeted therapy uses drugs that block the growth of breast cancer cells. Trastuzumab is one of the drugs being used in the therapy.

Fig. 5.1 shows the structure of HER2 protein and Trastuzumab.

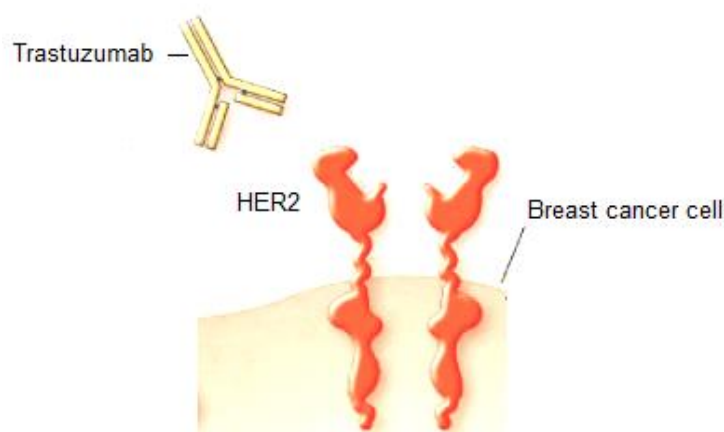


Fig. 5.1

(b) Suggest how amplification of *HER2* gene could result in breast cancer. [2]

1. Increases the number of copies of *HER2* gene;
2. therefore more HER2 receptors synthesised;
3. for growth factors to bind to;
4. stimulate uncontrolled cell division;

(c) Suggest how Trastuzumab works in cancer treatment.

[2]

1. shape complementary to the binding site of receptor;
2. block binding of growth factor;
3. prevent dimerization of receptor;
4. block signal transduction;
5. decrease cell division;

[Total: 9]

6.

Part I

In an investigation, animal cells were exposed to different concentrations of glucose. The rate of uptake of glucose into cells across the cell surface membrane was determined for each concentration. Fig 6.1 shows the results.

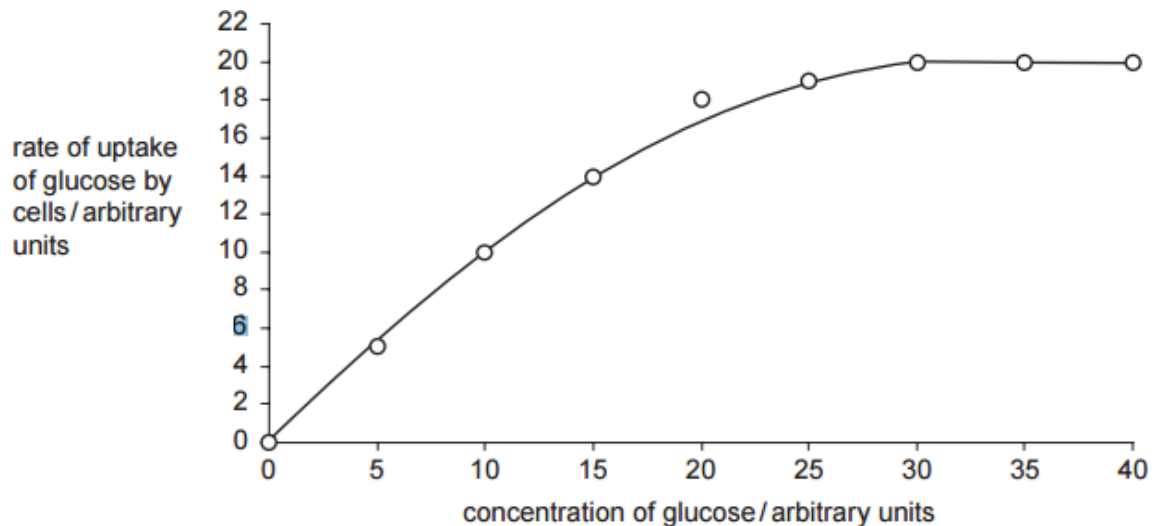


Fig 6.1

(a) Using the information from Fig 6.1,

(i) name the process by which glucose taken up into the cell. [1]

Facilitated diffusion;;

(ii) give reasons to support your answer in **(a)(i)**. [2]

1. the rate of uptake increases from 0 to 20 a.u. with increasing glucose concentration from 0 to 30 a.u.;
2. up to a plateau when glucose concentration is at 30 a.u. / constant rate at 2.0 a.u. from 30 to 40 a.u.;
3. all carrier/channel/transport proteins in use;

If glucose was taken up by

4. passive diffusion, rate would continue to rise;

Part II

Following digestion, glucose is transported via the bloodstream and taken up by animal cells such as liver cells. In an investigation, mammalian liver cells were homogenised and the resulting homogenate centrifuged.

Samples of the complete homogenate and samples containing only the mitochondria or only the cytosol were incubated with:

1. Glucose
2. Pyruvate
3. Glucose and cyanide
4. Pyruvate and cyanide

After incubation, the presence or absence of carbon dioxide and lactate in each sample was determined. The results are shown in Fig 6.2.

	samples of homogenate					
	complete		only mitochondria		only cytosol	
	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate
1 glucose	✓	✓	✗	✗	✗	✓
2 pyruvate	✓	✓	✓	✗	✗	✓
3 glucose and cyanide	✗	✓	✗	✗	✗	✓
4 pyruvate and cyanide	✗	✓	✗	✗	✗	✓

✗ = absent ✓ = present

Fig 6.2

- (b) Explain why carbon dioxide is produced when the sample with mitochondria only is incubated with pyruvate but not when it is incubated with glucose. [3]

1. Glycolytic enzymes is present in the cytosol;
2. Glycolysis does not occur in the mitochondria/can only occur in the cytosol;
3. Glucose not converted to pyruvate in sample with mitochondria only;
4. Pyruvate can enter mitochondrion while glucose cannot;
5. Carbon dioxide produced during link reaction;
6. and Krebs cycle;

In the presence of cyanide, lactate is produced but carbon dioxide is not. In addition, the oxygen level did not decrease.

(c) Suggest an explanation for the observation.

[4]

1. Ref to cyanide as an inhibitor of oxidative phosphorylation;
2. leads to members of the ETC to remain in the reduced state;
3. Oxygen not used as the final electron acceptor;
4. Reduced NAD not oxidized/ NAD not regenerated;
5. Krebs cycle/link reaction stops;
6. Pyruvate is reduced to;
7. Lactate produced in cytoplasm;
8. Via anaerobic respiration;
9. Involving lactate dehydrogenase;

Note: points 3 – 5 awarded only if cause and effect are correct

[Total: 10]

7. Coat colour in horses may be black, chestnut or gray. These colour differences are controlled by two independently assorting genes. In the absence of coloured pigments, the coat of the horse is gray.

A farmer, Zheng, crossed a pure-breeding stallion with chestnut coat and a pure-breeding mare with gray coat.

All the F_1 offsprings were horses with gray coat. The offsprings were allowed to interbreed and the results of the cross were recorded according to the coat colour of the horses:

Horses with gray coat	130
Horses with black coat	34
Horses with chestnut coat	12

- (a) Using the symbols below, draw a genetic diagram to show the cross described. [4]

G – Gray **g** – Coloured **B** – Black **b** – Chestnut

Parental phenotypes: Gray Coat x Chestnut Coat
 Parental genotypes: **GGBB** x **ggbb**;
 Gametes produced by parents: **(GB)** ; **(gb)** ;
 F_1 genotypes: **GgBb**
 F_1 phenotypes: All gray
 Interbreeding of F_1 generation: **GgBb** x **GgBb**
 Gametes produced: **(GB)** **(Gb)** **(gB)** **(gb)** ; **(GB)** **(Gb)** **(gB)** **(gb)** ;
 Punnett Square to show the fusion of gametes produced: (genotype;; phenotype;;)

Genotype;; Phenotype;;	♀ \ ♂	(GB)	(Gb)	(gB)	(gb)
	(GB)	GGBB ● Gray	GGBb ● Gray	gGBB ● Gray	GgBb ● Gray
	(Gb)	GGBb ● Gray	GGbb ● Gray	GgBb ● Gray	Ggbb ● Gray
	(gB)	GgBB ● Gray	GgBb ● Gray	ggBB ▲ Black	ggBb ▲ Black
	(gb)	GgBb ● White	Ggbb ● Gray	ggBb ▲ Black	ggbb ■ Chestnut

F_2 phenotypic ratio: 12 Gray : 3 Brown : 1 Chestnut

- (b) Explain the interaction between the gene loci that leads to the ratio observed in (a).

[4]

1. Dominant epistasis;
2. Dominant allele *G* is epistatic to gene locus B/b;
3. Allele *G* codes for inhibitor;
4. which inhibits enzyme which converts gray pigment precursor to chestnut pigment;
5. When the horse has the genotype *gg*/in the absence of allele *G*, the phenotypic expression of gene B/b is not prevented;
6. thus horses will have coloured/chestnut and brown coat;
7. When the horse has the genotype *ggb*, the proteins/enzymes involved in coloured pigment synthesis are not synthesized/non-functional;
8. thus coat colour is chestnut;
- or
9. Allele B determine black coat;
10. While allele b determines chestnut coat;

- (c) A chi-squared test was performed on the results of the cross, and the calculated χ^2 value obtained is 0.152.

Table 7.1

Degrees of freedom	probability				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Using the χ^2 value provided and Table 7.1, describe what conclusion may be drawn.

[2]

1. Calculated χ^2 value corresponds to p value greater than 0.05 / $p > 0.10$ which is > 0.05 ;
2. do not reject null hypothesis;
3. the difference between the observed and expected results is statistically insignificant;
4. and difference between the observed and expected is due to chance;

[Total: 10]

8. The Death Valley region of Nevada in the USA used to have an extensive lake system. Approximately 20 000 years ago the lakes started to dry up and now consist of isolated small pools. Four different species of the desert pupfish have been found living in these pools.

Evidence indicates that over 20 000 years ago there was only one species of pupfish living in the lake system.

Fig. 8.1 shows the lake 20 000 years ago and a common pupfish found in the lake.

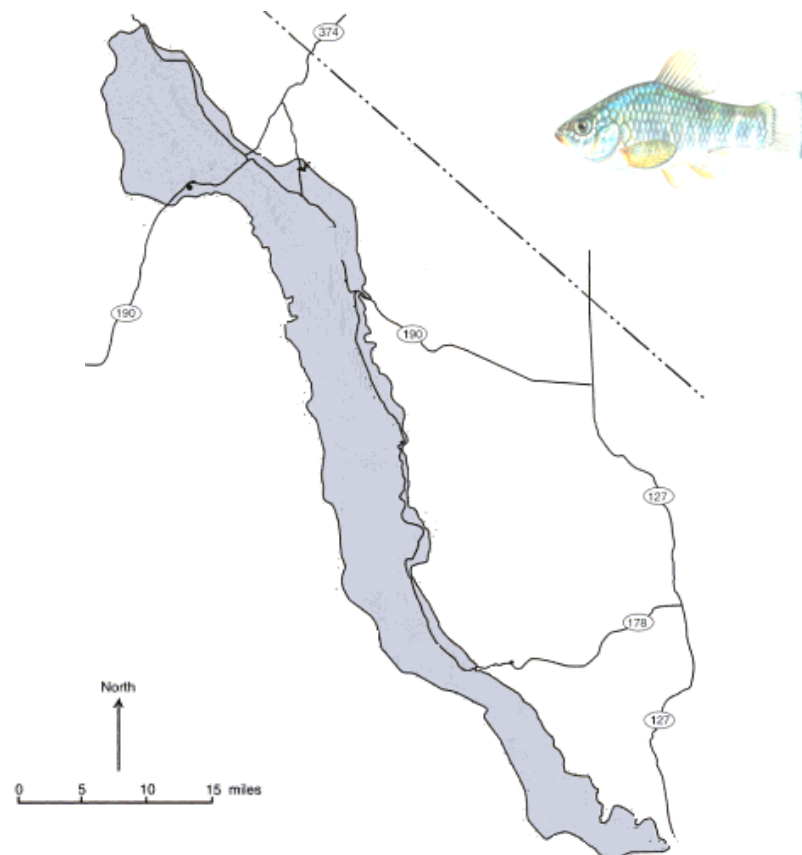


Fig. 8.1

(a) Explain how the change from an extensive lake system to just a few pools could have resulted in the evolution of four new species of desert pupfish. [4]

1. phenotypic variation due to mutations occur;
2. different pools exert different selection pressures / different environmental conditions;
3. favouring different phenotypes;
4. individuals with favourable phenotype, survive and reproduce, pass on the allele to the offspring;
5. change in allele frequency;
6. geographical isolation of fish populations;
7. no interbreeding and no gene flow between populations;
8. allopatric speciation;
9. genetic drift;

(b) State how environmental factors can act as stabilising forces of natural selection in an isolated pool, after the initial evolution of a new species of desert pupfish. [2]

1. conditions remain the same within the pool;;
2. best adapted fish to conditions in pool survive;
3. extreme phenotypes are selected against / do not survive;

Fig. 8.2 shows the nucleotide base sequence of a length of DNA from the gene coding for cytochrome b from the four pupfish species. The nucleotide bases with no change with reference to species 1 are indicated by a dot.

Species	Nucleotide base																			
1	G	G	A	T	G	A	T	A	T	G	G	G	G	A	T	G	T	T	G	C
2	.	.	.	G	.	.	.	.	.	.	.	A	.	A	.	.	A	.	.	.
3	C	A	G	C	.	.	.	.	.	.	A	A	A	.	.	.	A	.	G	A
4	.	.	.	G	.	.	G	.	.	.	.	A	.	A	.	.	A	.	.	T

Fig. 8.2

- (c) (i) Suggest why the cytochrome b gene is used to measure changes in DNA sequences in closely related species. [2]
1. the gene was present in all of the species so a good basis for comparison;;
 2. the gene is found in mitochondria and passed down the maternal line;;
- (ii) Explain how the nucleotide sequences in Fig 8.2 can be used to determine the evolutionary relationships between the four pupfish species. [2]
1. the smaller the number of differences the, more closely related / more recently divergence occurred;;
 2. data quote;;

[Total: 10]

Section B (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

9. (a) Compare and contrast collagen and haemoglobin. [7]
- (b) Define control elements and explain how they influence transcription. [7]
- (c) Describe mode of action of DNA polymerase. [6]

[Total: 20]

9. (a) Compare and contrast collagen and haemoglobin.

[7]

Similarities

- S1. Monomer: amino acid;;
- S2. Linked by peptide bond;;
- S3. Consists of more than 1 subunit;;
- S4. AVP;;

Differences [Max 5]

	Feature	Collagen	Haemoglobin
D1	shape	Fibrous	Globular
D2	Types of amino acids	Mostly made up of glycine, proline and hydroxylysine or hydroxyproline	Made up of a variety of amino acids
D3	Sequence of amino acids	a regular repetitive sequence	No fixed sequence
D4	Level of organization	Secondary structure & Quaternary structure	Tertiary structure & Quaternary structure
D5	Molecular structure	3 polypeptide chains wind around each other to form triple helix	4 polypeptide chains, each polypeptide chain tightly folded to form spherical shape
D6	Bonds	Cross-linked by hydrogen and covalent bonds	Interaction of the 4 types of bonds: ionic, hydrogen, disulfide and hydrophobic interactions <i>Reject: R group interactions</i>
D7	Length of polypeptide chain	Each polypeptide chain comprises of about 1000 amino acid residues	Each polypeptide chain contains fewer numbers of amino acids (α -chain $\rightarrow$ 141 amino acids; β -chain $\rightarrow$ 146 amino acids)
D8	Prosthetic group	Absence of prosthetic group	Presence of prosthetic group (haem) with protein
D9	Function	Structural support	Transport of oxygen
D10	Solubility in water	Insoluble in water	Soluble in water

1 m for each comparison

(b) Discuss how control elements influence transcription.

[7]

Control element

1. non-coding DNA where transcription factors bind;
2. to regulate gene expression;
3. control elements consists of promoter, enhancer and silencer;

Promoter

4. consist of the TATA box and the transcriptional start site;
5. TATA box serves as the binding site of general transcription factors; that
6. facilitate the binding of RNA polymerase to the transcription start site;
7. determines basal level of transcription;

Proximal promoter

8. CAAT boxes and GC-rich sequences upstream from promoter;
9. serves as sites that bind other regulatory proteins.
10. determine how frequent the transcription event occurs

Enhancer

11. attachment of activators to enhancers;
12. increases the rate of transcription;
13. to recruit DNA bending protein;
14. to help RNA polymerase and transcription factors assemble at the promoter / contact each other more tightly forming transcription initiation complex;

Silencers

15. attachment of repressors to silencers;
16. decreases the rate of transcription;
17. prevent the activators from binding to enhancer;
18. bind to activation domain of activator;
19. interferes with formation of the initiation complex;

(c) Describe mode of action of enzymes, such as DNA polymerase. [6]

1. Enzymes lowers the activation energy;
2. enzyme holds the substrates in an arrangement that forces them in the correct orientation and in close proximity;
3. increases probability of a reaction occurring;

Chemical fit

4. Enzyme and substrate must have charge and hydrophilic/hydrophobic complementarity;

Spatial fit

5. only substrate of complementary shape will fit into the active site;
6. binding of substrates to enzyme active site forming enzyme substrate complexes;
7. Ref to lock & key hypothesis - the substrate is the 'key' and enzyme's active site is the 'lock';;
8. Ref induced fit hypothesis - binding of substrates to active site induces a conformational change enables the substrate to fit more snugly into the active site;;

Specificity

9. DNA polymerases are highly specific in the reactions they catalyse;
10. DNA polymerase is only able to add nucleotides to an 3'-OH group of the last nucleotide in the growing strand due to the specificity of its active site;;
11. resulting in the formation of phosphoester bonds between nucleotides of DNA strand;;

[Total: 20]

10. (a) Explain the sentence 'The expression of lactose metabolising proteins in *E. coli* is under positive and negative gene regulation'. [8]
- (b) Describes the role of insulin in the regulation of blood glucose level in human. [8]
- (c) Outline the advantages of signal transduction. [4]

10. (a) Explain the sentence 'The expression of lactose metabolising proteins in *E. coli* is under positive and negative gene regulation'. [8]

1. Genes encoding lactose metabolizing proteins are clustered together in *lac* operon;
2. *LacI*, encodes an active repressor;

In negative gene regulation [max 4]

3. In the absence of lactose, *lac* repressor binds to the operator;
4. *lac* operon is switched off;
5. In the presence of lactose, lactose is converted to allolactose by β -galactosidase;
6. Allolactose binds to allosteric site of *lac* repressor protein;
7. Resulting in a change in three dimensional conformation of the *lac* repressor protein;
8. Causing *lac* repressor to adopt its inactive conformation;
9. *lac* repressor protein can no longer bind to the operator;
10. This switches on the *lac* operon;
11. Allowing for RNA polymerase to bind to the promoter;
12. Structural genes of the *lac* operon are transcribed;
13. Enzymes for lactose metabolism (β -galactosidase, permease, transacetylase) are synthesized;

In positive gene regulation [Max 3]

14. *E. coli* preferentially utilize glucose over lactose;
15. When glucose concentration is low, cAMP level is high;
16. cAMP binds to Catabolite Activator Protein (CAP);
17. leads to a change in three dimensional conformation of CAP;
18. active CAP binds to the promoter of the *lac* operon;
19. and interact with RNA polymerase/ increase the affinity of RNA polymerase to the promoter;

20. Increasing the rate of transcription of structural genes;
21. More enzymes for lactose metabolism (β -galactosidase, permease, transcetylase) are synthesized;

(b) Describe the role of insulin in the regulation of blood glucose level in human [8]

1. High blood glucose concentration above normal level/set/reference point/ 90mg/100ml is detected by β -cells;
2. of the islets of Langerhans;
3. β -cells secretes insulin into the bloodstream ;
4. insulin will bind to insulin receptors on liver/target/effector cells;
5. hormone-receptor complexes dimerises;
6. and autophosphorylate
7. activated/phosphorylated receptor acts as a kinase
8. phosphorylating intracellular relay protein known as insulin receptor substrate (IRS)
9. phosphorylated IRS recognises and binds various intracellular proteins at SH2 domains
10. initiate a cascade of phosphorylation
11. leading to activation of glycogen synthetase
12. for glycogenesis
13. activation of glycolytic enzymes, e.g. glucokinase,
14. for increased rate of glycolysis;
15. inhibition of enzymes involved in glycogenolysis;
16. inhibit gluconeogenesis;
17. Increase glucose uptake into liver and muscle cells;
18. resulting in decrease in blood glucose level, back to the norm;

(c) Outline the advantages of signal transduction.

[4]

1. Signal amplification;
2. molecules in a pathway transmit the signal to numerous molecules at the next step in the series;
3. this result can be a large number of activated molecules;
4. Greater opportunity for coordination and regulation / fine-tuning of the response;
5. Multistep pathways afford greater opportunity for coordination and regulation compared to simpler systems;
6. Relay system;
7. A ligand binding to specific receptors at the cell membrane can lead to corresponding cellular response;
8. Simultaneous activation of many cells;
9. a ligand can activate several processes at the same time;